

GENERAL INFORMATION

ABOUT THIS MANUAL [G2255554]

DESCRIPTION AND OPERATION

INTRODUCTION

The workshop manual has been written in a format that is designed to meet the needs of technicians worldwide. The objective is to provide descriptions for completing diagnosis and testing, service and repair work with tested and effective techniques, using a common publishing format.

It is structured into **groups** which are specific to vehicle areas, **sections** which are component areas, and **sub sections** which contain lists detailing Specifications, Description and Operation, Diagnosis and Testing, General Procedures, Disassembly and Assembly, Removal and Installation.

HOW TO USE THIS WORKSHOP MANUAL

Appropriate service methods and repair procedures are essential for the safe, reliable operation of all vehicles as well as the personal safety of the individual carrying out the work.

Anyone who does not follow the instructions provided in this manual, must first establish that personal safety or vehicle integrity is not compromised by the choice of method, tools or components.

WARNINGS, CAUTIONS AND NOTES

WARNING:

Warnings indicate hazards that may be present while carrying out the procedure. These hazards may cause personal injury if not followed.

CAUTION:

Cautions indicate that failure to follow the instruction may result in damage to the vehicle or equipment.

NOTE:

Notes provide additional information that is required in the procedure.

General warnings, cautions and notes are included before any procedural steps. These are only used when they apply to each step contained in the procedure.

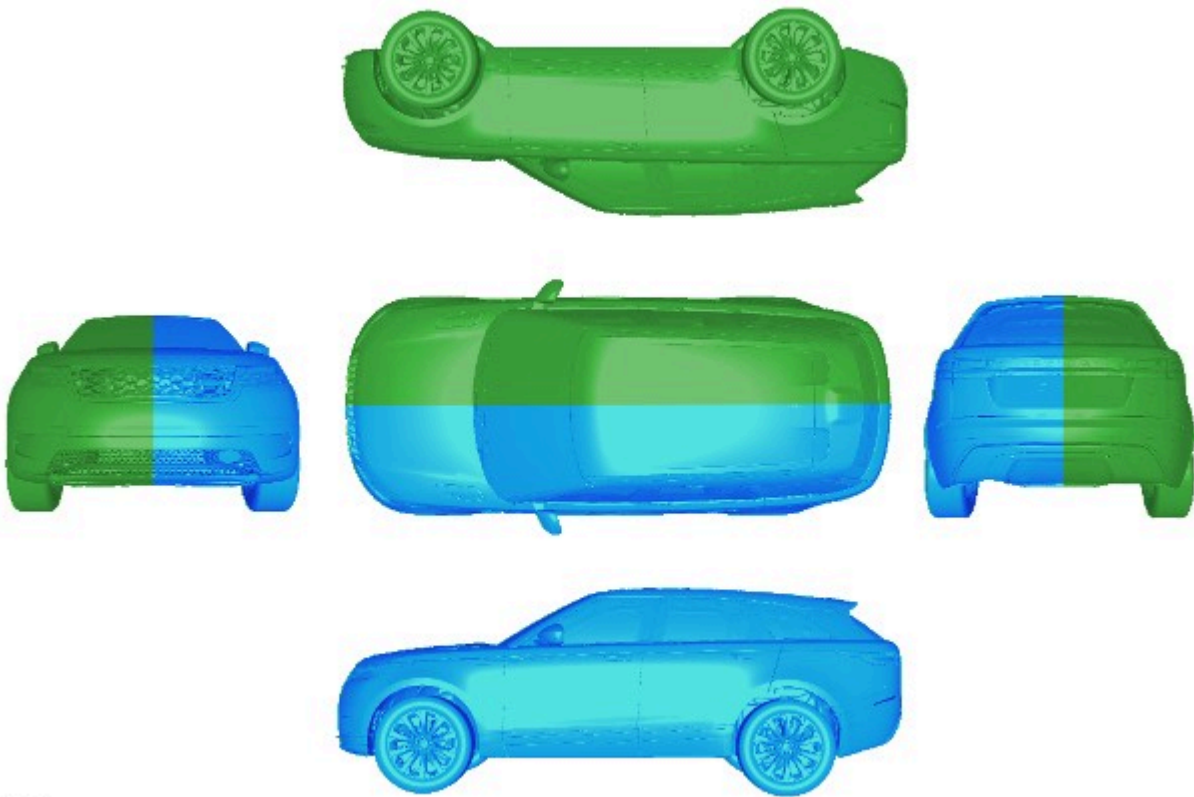
Step specific warnings, cautions and notes are only assigned to the individual step that they apply to.

POSITIONAL REFERENCES

Vehicle

Positional references of the vehicle are always from the drivers seat looking forward.

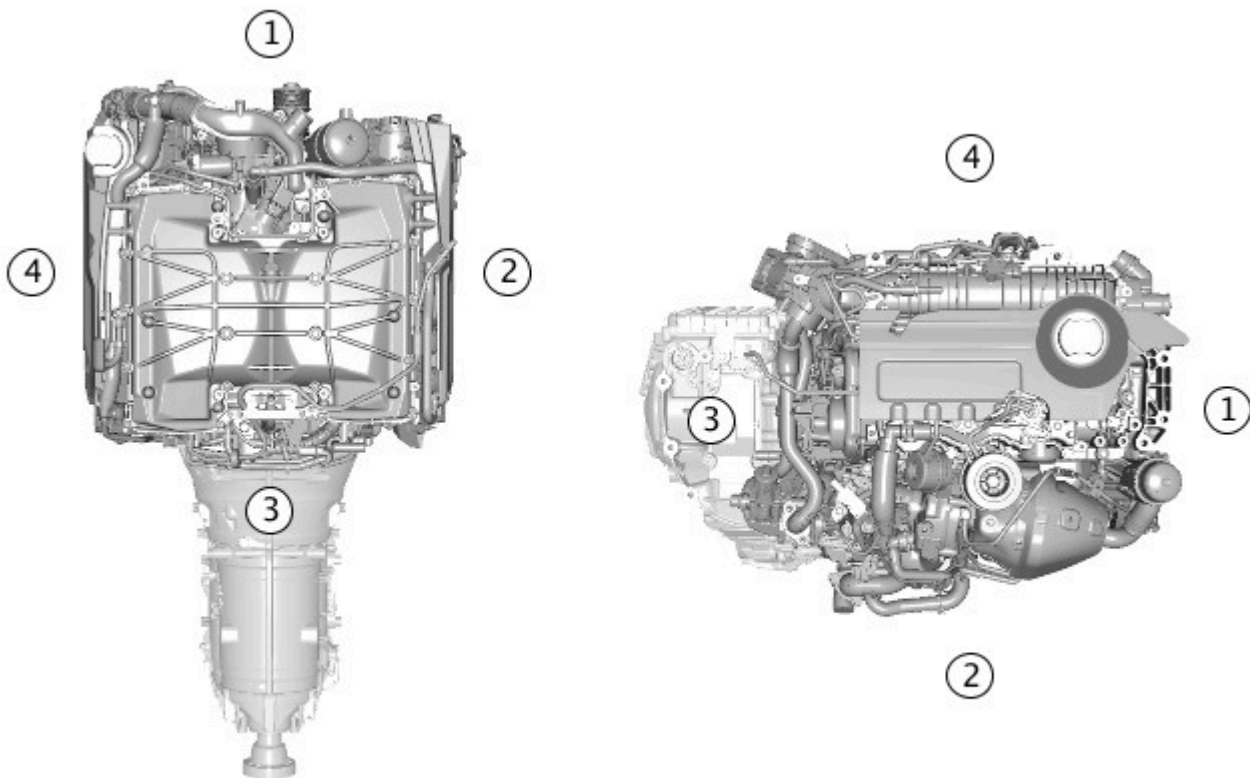
Left side of the vehicle is shown blue, right side of the vehicle is shown green.



E222998

Engine

Positional references of the engine are always from the flywheel, or drive plate, looking towards the crankshaft pulley.



E222999

ITEM	DESCRIPTION
1	Front
2	Right
3	Rear
4	Left

PROCEDURE AUTHORIZING STANDARDS

A storyboard style procedure is the preferred standard used when creating the following types of procedures:

- General Procedures
- Removal
- Installation
- Removal and Installation

The procedures describe the removal and/or installation required to access the serviceable part.

Components or assemblies may be removed and/or installed to access other components or assemblies. In this case, no further disassembly and assembly is required and the procedure steps will have a statement, displayed as a note, providing this information.

Special Tools

If special tools are required to perform the procedure, these will be displayed in a table showing an image of the special tool, the special tool number and a description.

SPECIAL TOOL(S)

	204-269 Flange remover forcing screw		205-053 Retainer, Differential Pinion Flange
	205-053-03 Adapter, Differential Pinion Flange		307-520 Installer, Output Shaft Seal

E223000

Special tool numbers will also be referenced in all step illustrations where the special tool is visible in the active colour. Once the tool has been referenced it does will not be replicated in further steps.

General Equipment

If any general equipment is required to perform the procedure, these will be displayed in a table after the special tool table, if applicable, and before any procedural steps. The general equipment table includes all dimensionally specific items, such as drill bits. Tools that are used in an uncharacteristic fashion, particularly hand tools that would not be intuitive to perform the procedure will also be displayed.

GENERAL EQUIPMENT

EQUIPMENT NAME
6 mm drill bit
Transmission jack

E223001

New Parts

If new parts are required in any procedural step, these will be referenced in the parts table. There will be a reference to which step the new part is associated with.

PART(S)

STEP	REPLACE PART / RENEW PART	PART NAME
Removal Step 11	Replace Part	Lower timing cover
Removal Step 14	Replace Part	Lower timing cover

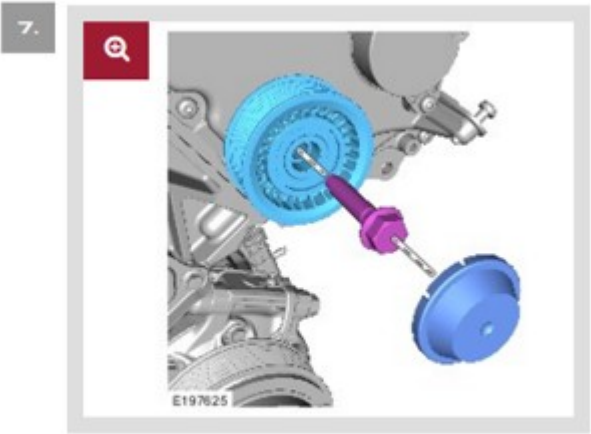
E223002

The general equipment table does not include hand tools such as wrenches, pliers and screwdrivers that are used as expected.

Step Types

All steps that include an illustration can have supporting text that contains a primary instruction. The illustrated step can contain supporting sub-steps where further clarification is necessary.

Basic Step - These steps can be supported with a primary instruction.



Remove the left side idler pulley.

E223004

Bullet Point Step - These is used for steps containing multiple actions/instructions in relation to the same component.

5.

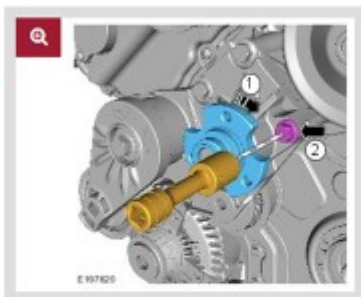


- Position the DTI probe 5 mm from the inner edge of the disc.
- Zero DTI and rotate road wheel one complete revolution to measure disc runout.

E223003

Numbered Sub-Step - These are used when specific order must be followed and are indicated in the illustration. Numbers within a illustration indicate the order of the steps and align with the supporting text.

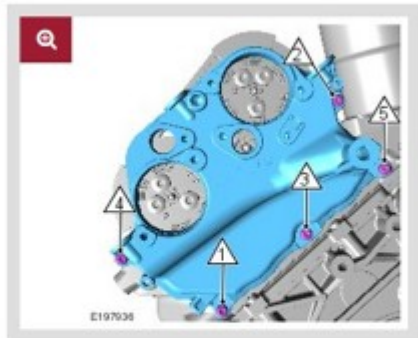
6.



1. Position the supercharger belt pulley shaft, as illustrated.
2. Remove the accessory drive belt pulley bracket bolt.

E223005

Sequence Step - There are used sequences, stages and/or loosening steps are shown in the procedure step. In most cases these will only be required in the installation steps.



- Install the right side upper timing cover.
- Tighten the bolts in the indicated sequence.

Torque: **12 Nm**

E223006

PROCEDURE ILLUSTRATION STANDARDS

All illustrations will have certain components and/or assemblies shown in specific colors. These colors are used to show different actions to provide further clarity.

NOTE:

At the start of 2018, a new color palette was introduced to further enhance the workshop manual.

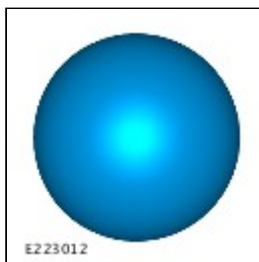
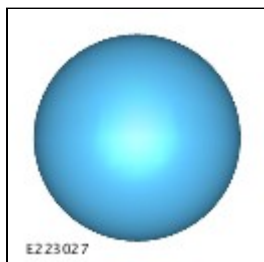
The definition of all the colors used are as follows:

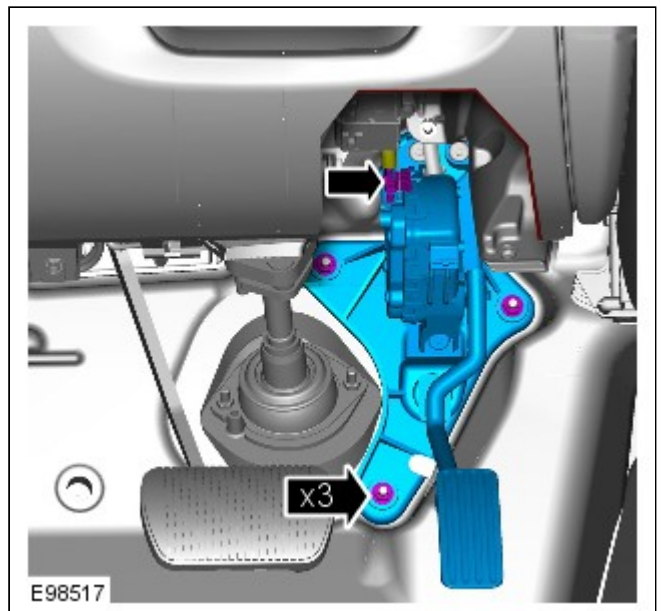
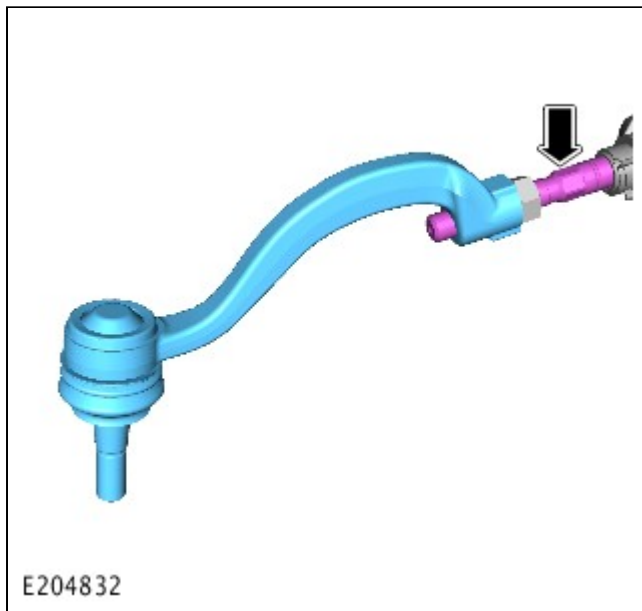
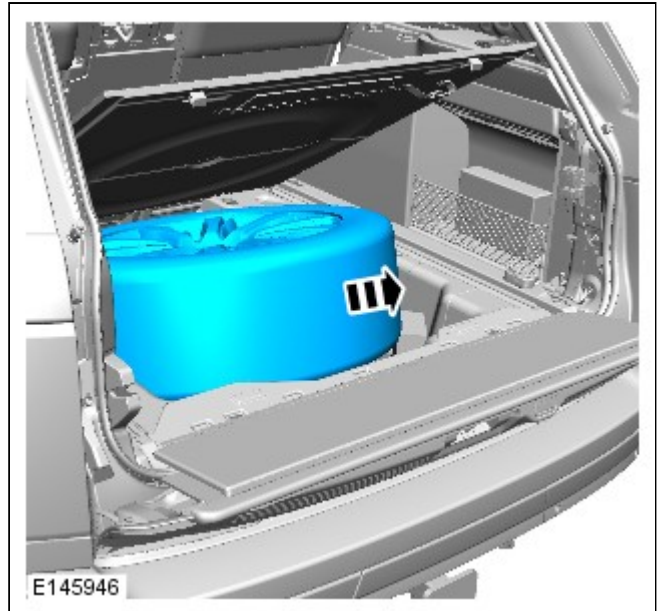
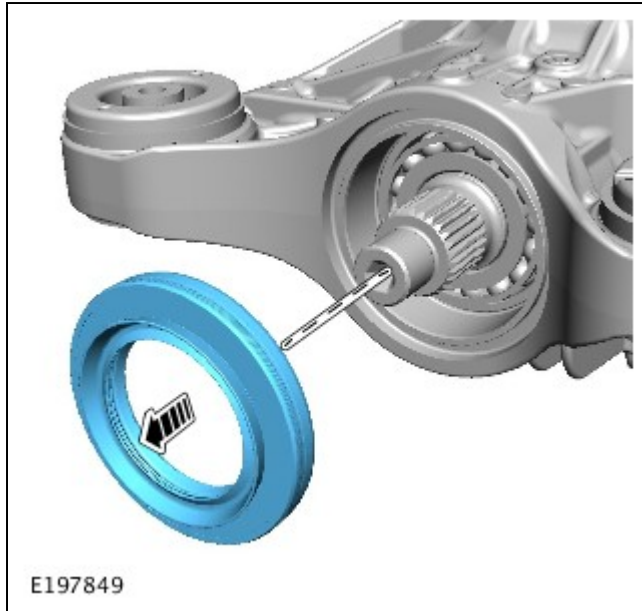
Focus Item - Main

A component to be removed/installed or disassembled/assembled. This color can also be used to highlight a feature/area such as timing or alignment mark, or regions to clean, grease or oil.

2018 ONWARDS

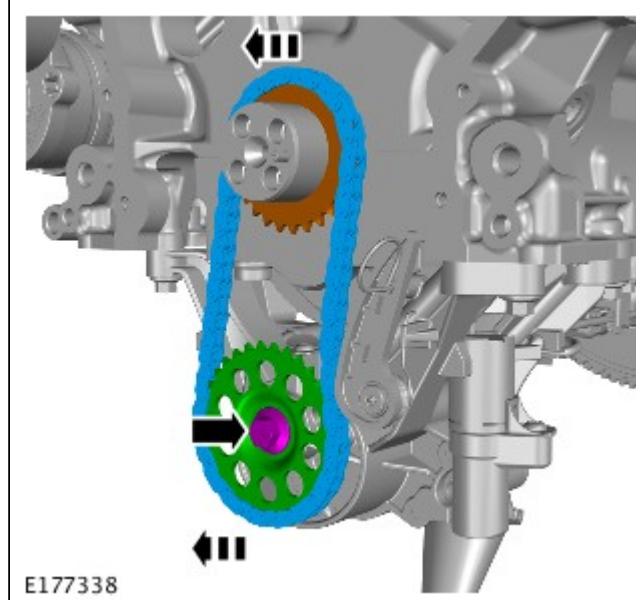
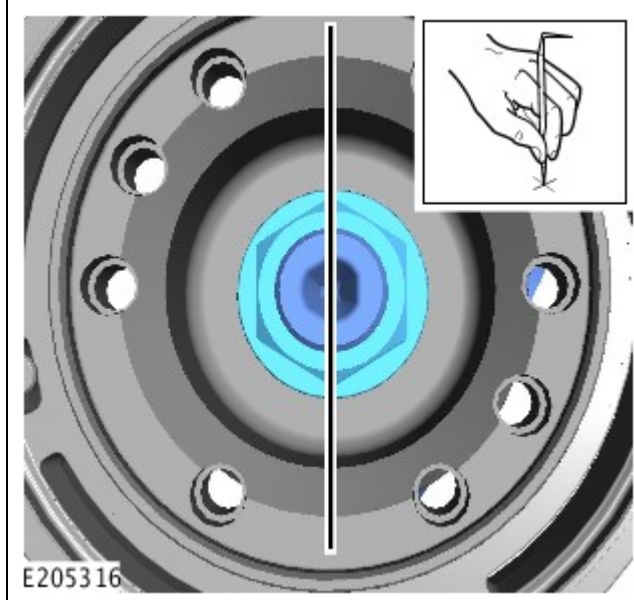
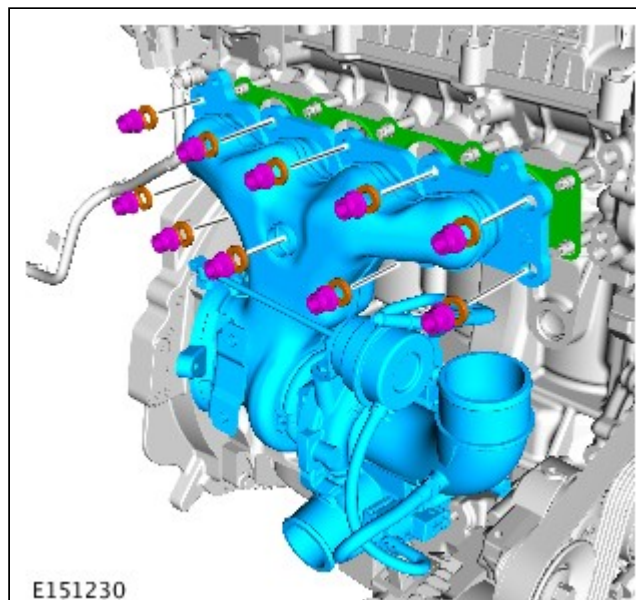
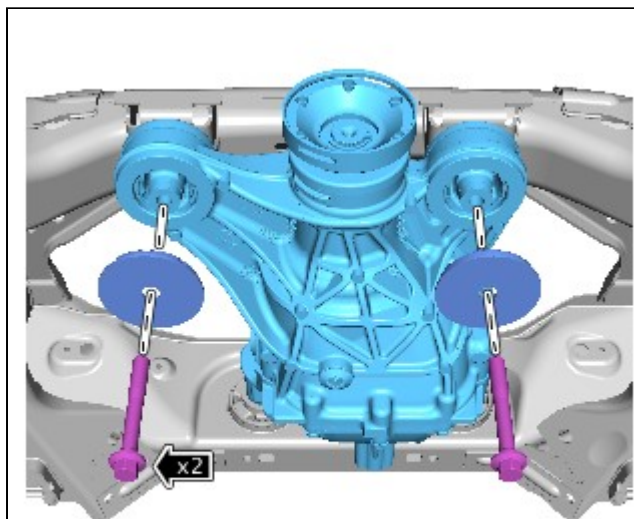
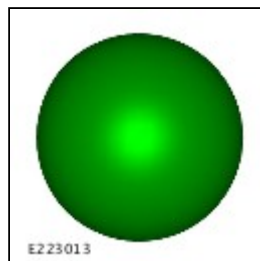
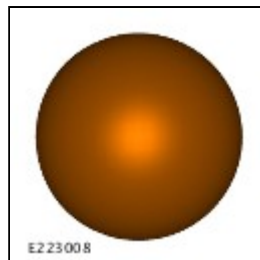
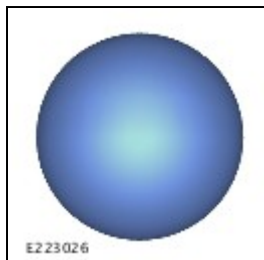
PRE 2018





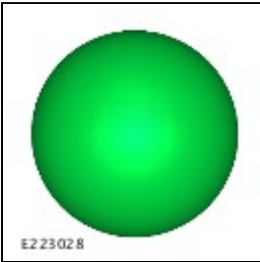
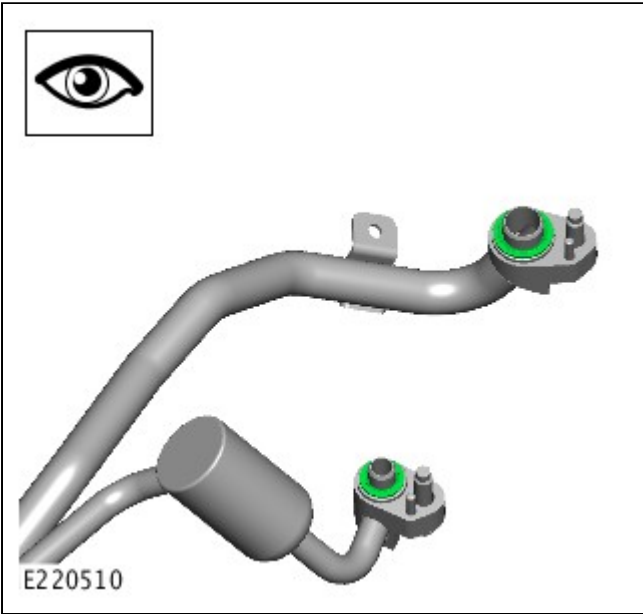
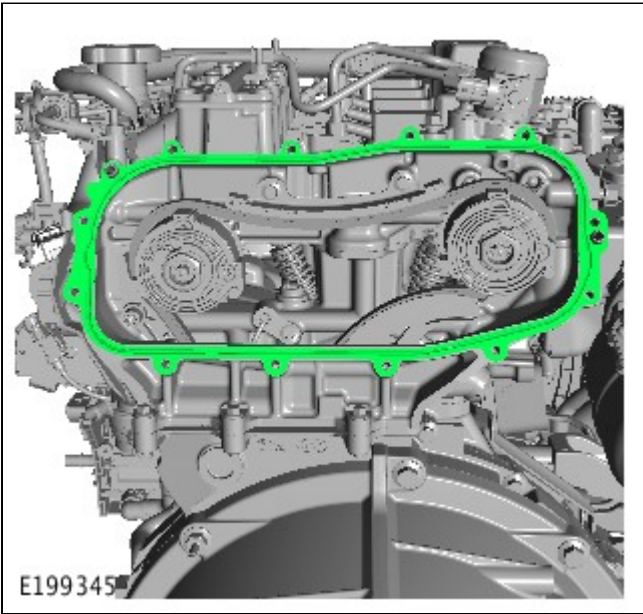
Focus Item - Alternative

An alternative color when multiple parts are to be removed in succession. This color, in addition to focus blue, is clearer to see how many parts are to be removed.



Gaskets and O-Ring Seals

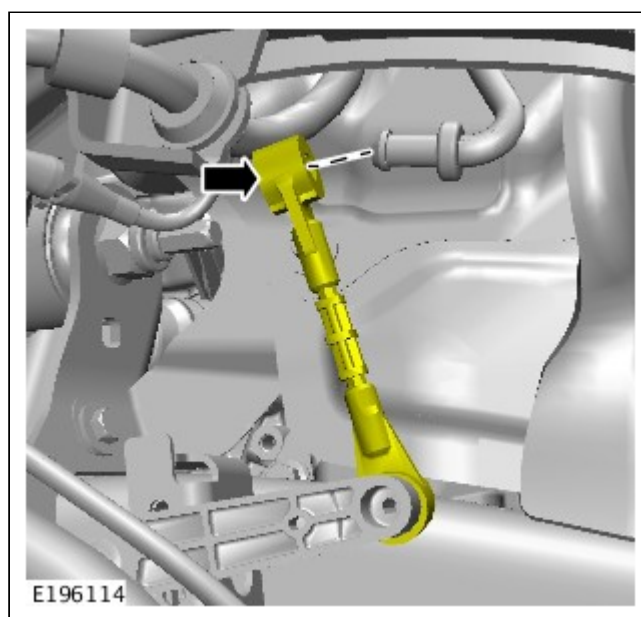
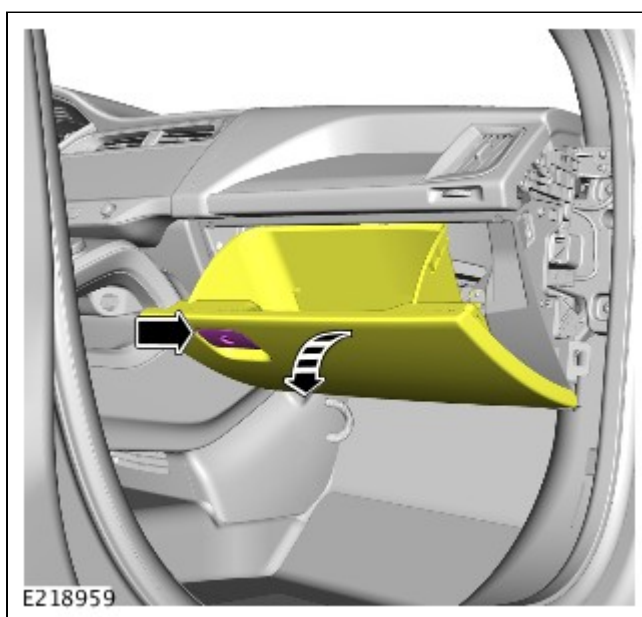
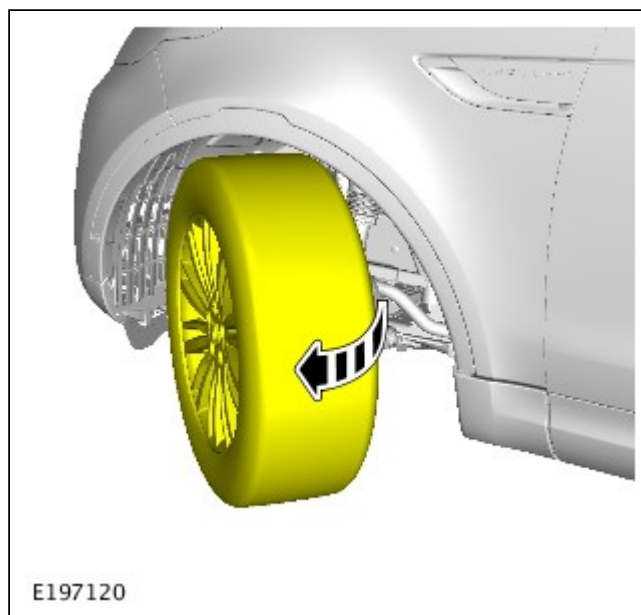
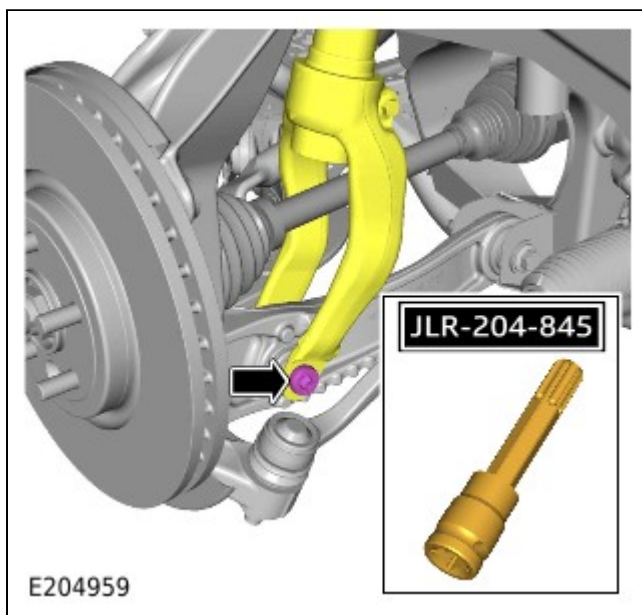
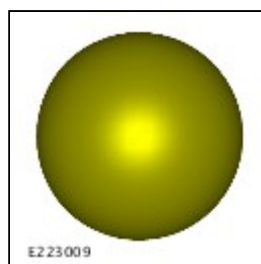
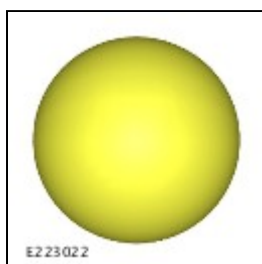
All gaskets and o-ring seals are shown in this color. If they are required to be replaced or inspected this will be detailed in the step.

2018 ONWARDS	PRE 2018
	Not Applicable
	Not Applicable
	Not Applicable

This color is used on components that are not being removed from the vehicle. These components are being released from another component, but remain connected or attached to the vehicle. It is also used to show the manipulation or aside of a component during the removal of another.

2018 ONWARDS

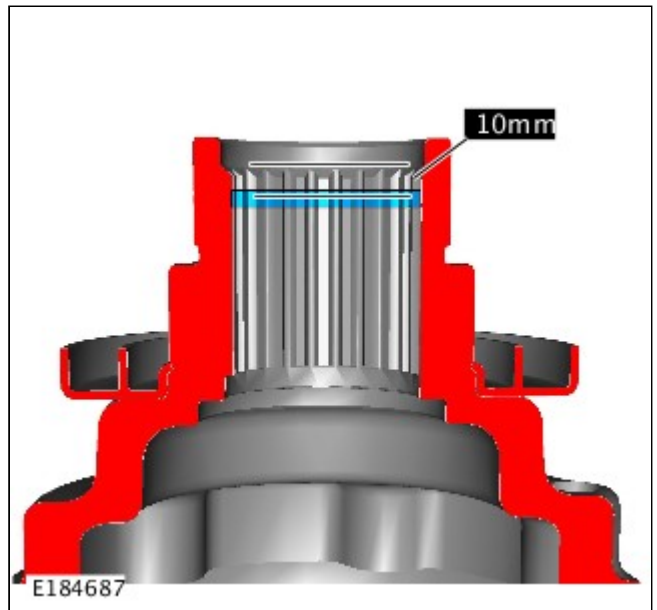
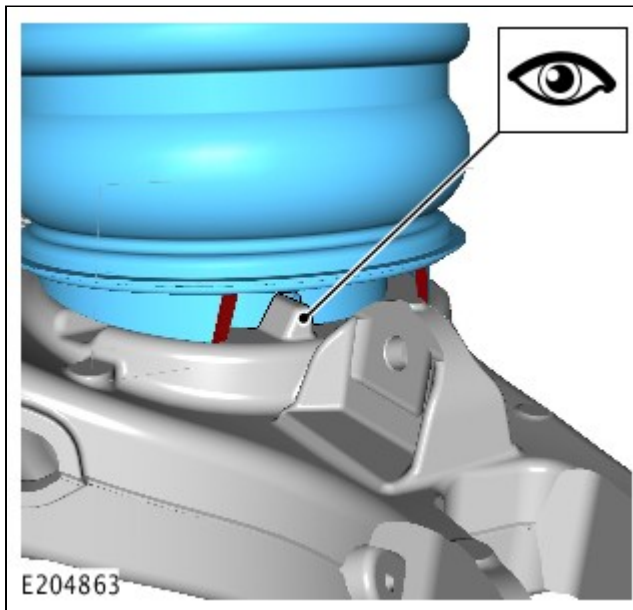
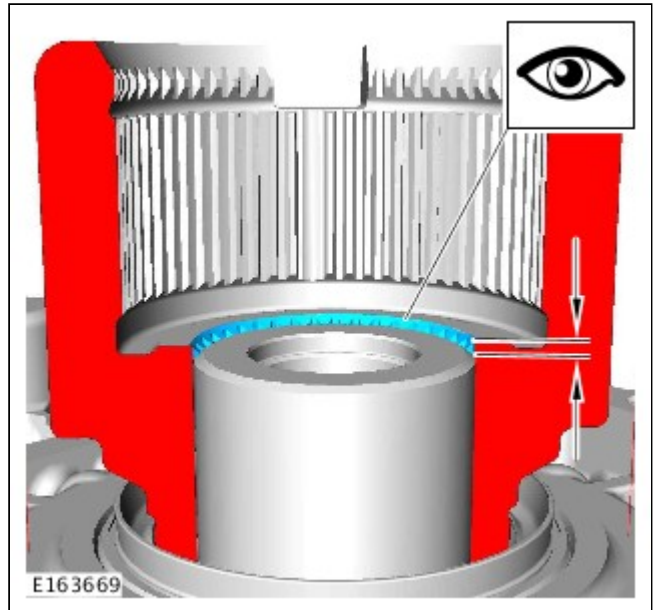
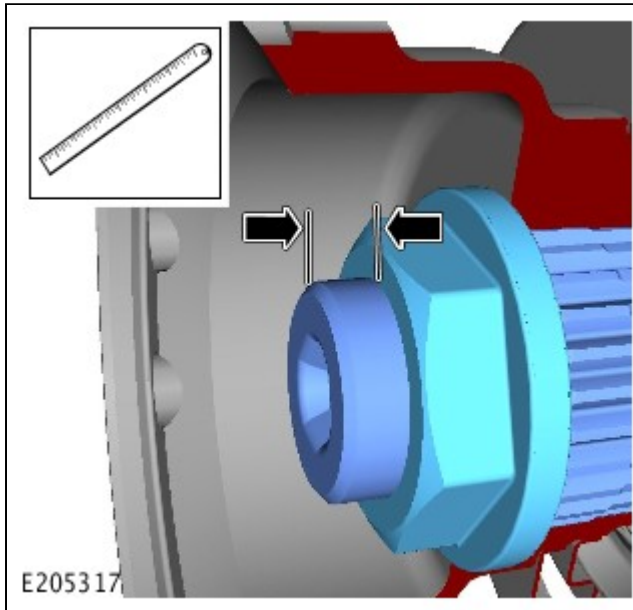
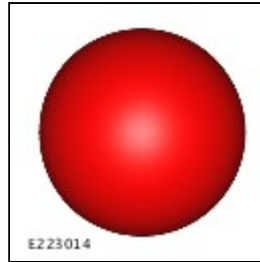
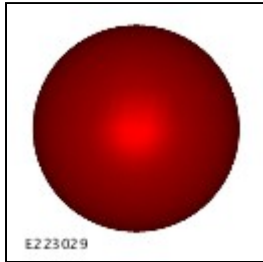
PRE 2018



Section capping is used as the component face color when a cross section is required. Black cross hatching is can be used depending on the detail of the components in question.

2018 ONWARDS

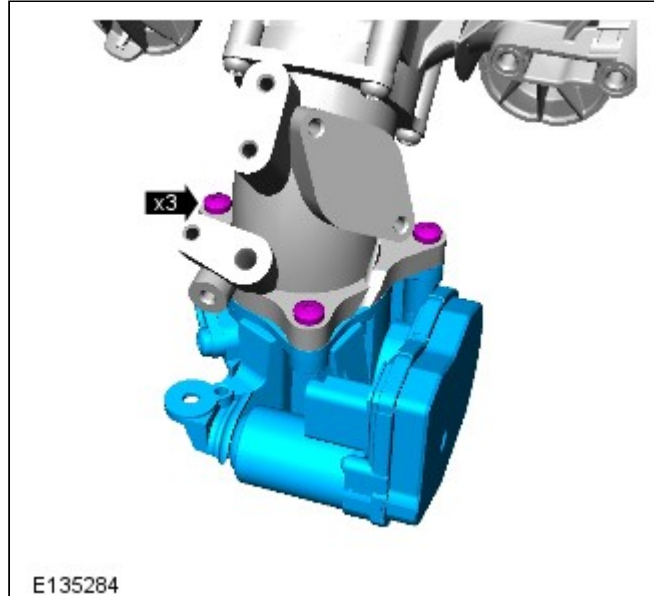
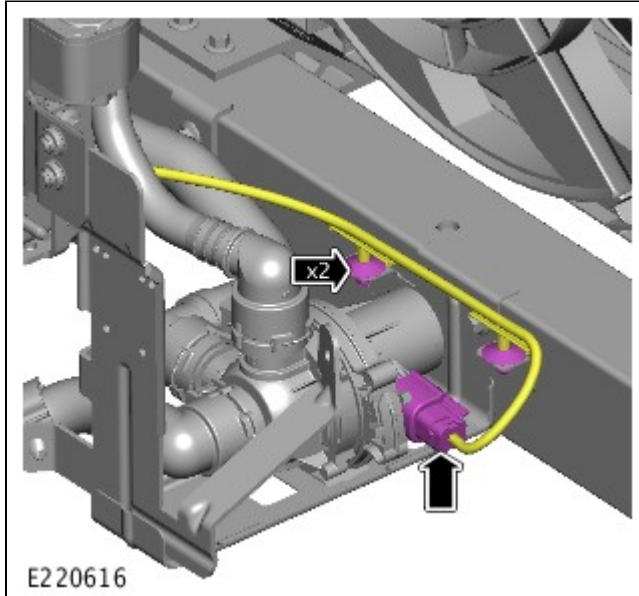
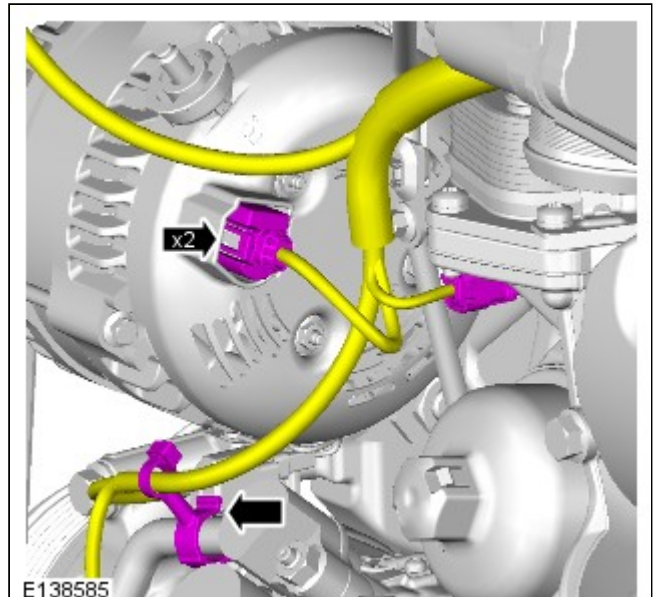
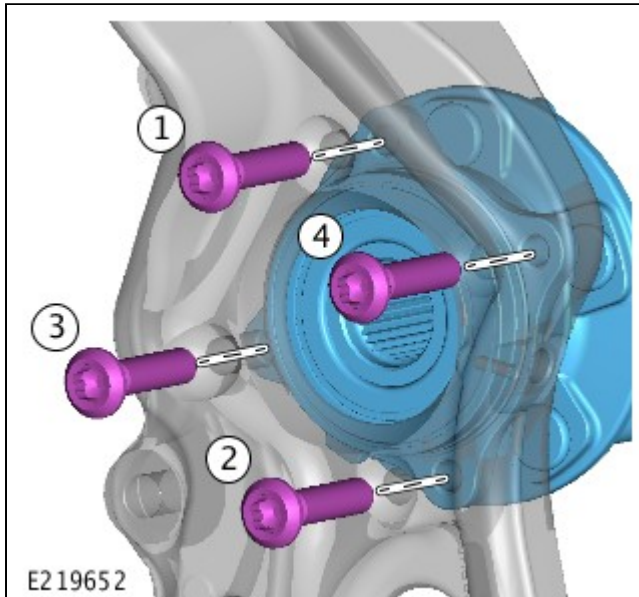
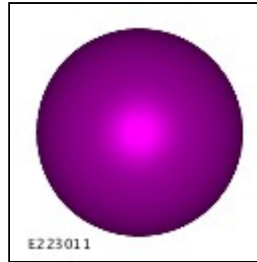
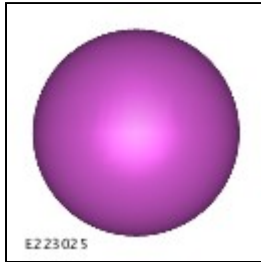
PRE 2018



This color is used on component that secures another by means of a thread, friction fit, or another form of lock/latching mechanism.

2018 ONWARDS

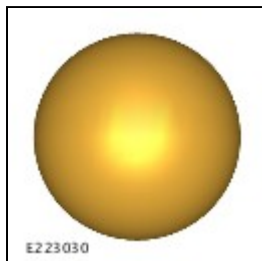
PRE 2018



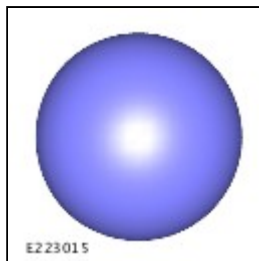
This color is used for all special tools and general equipment when installed or removed as a single entity. This special tool color 'active' represents a tool been positioned or manipulated during the step.

2018 ONWARDS

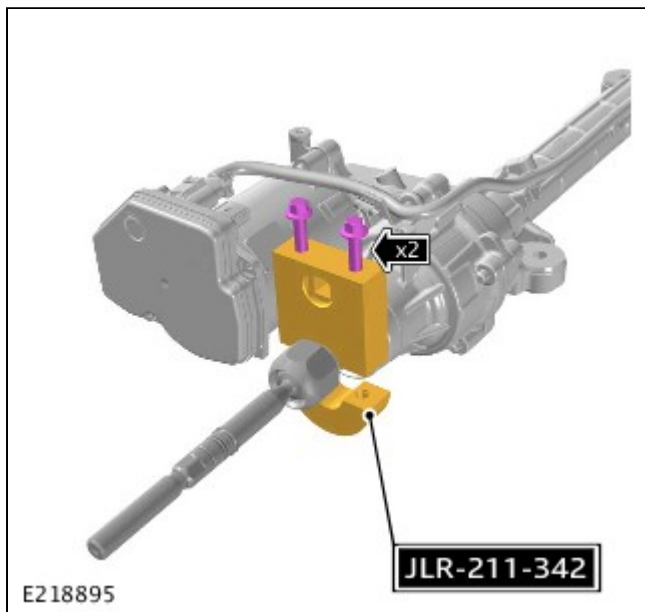
PRE 2018



E223030

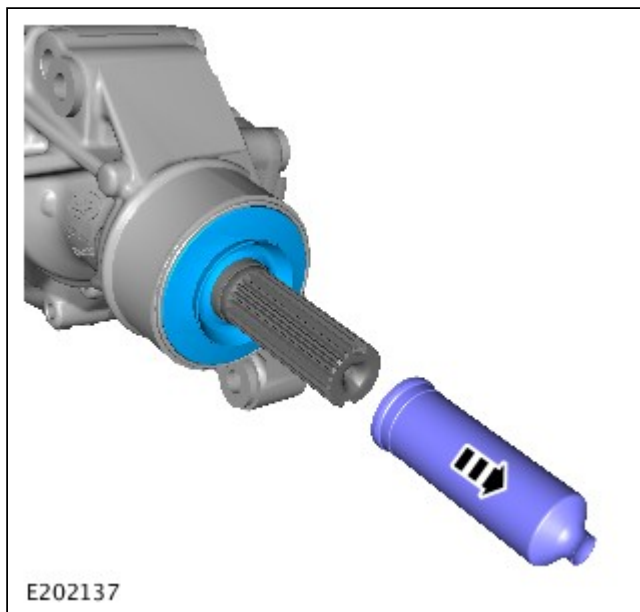


E223015

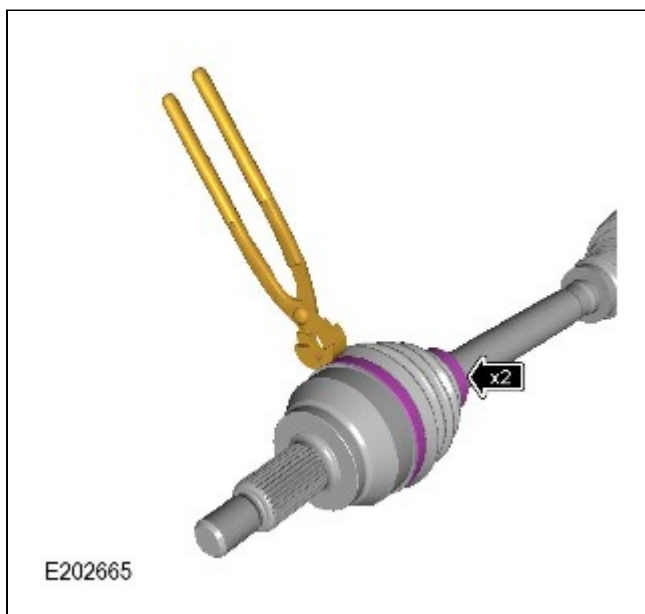


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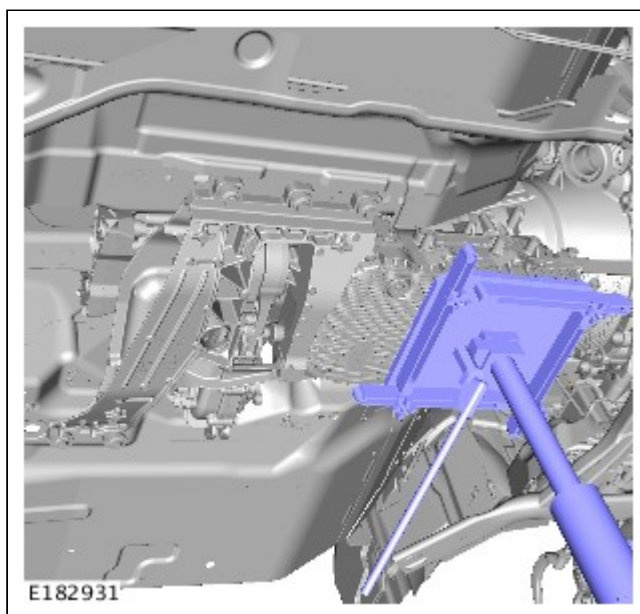
JLR-211-342



E202137

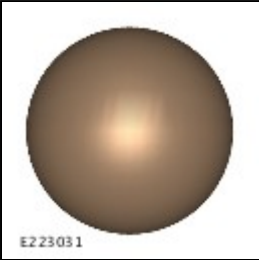
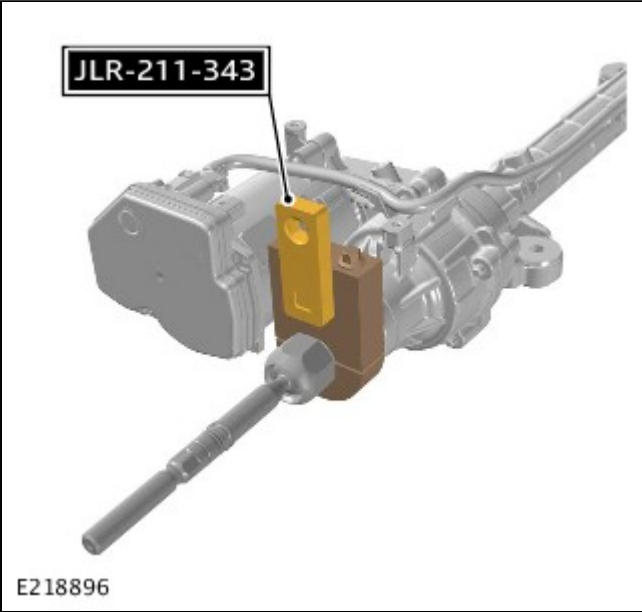
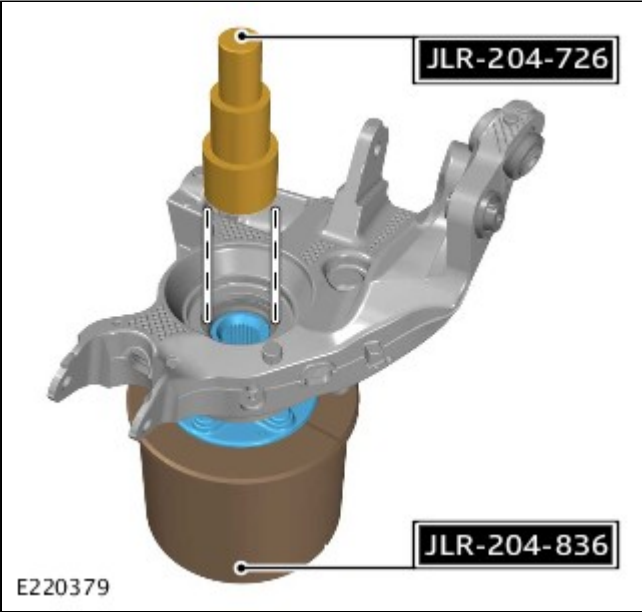


E202665



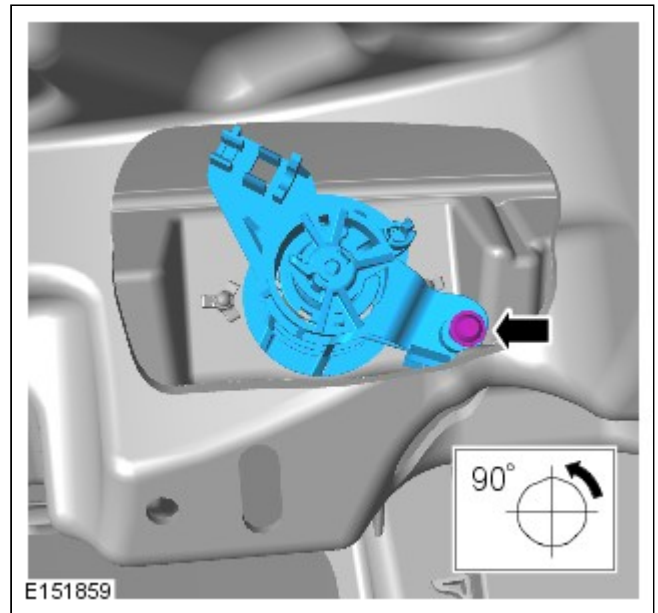
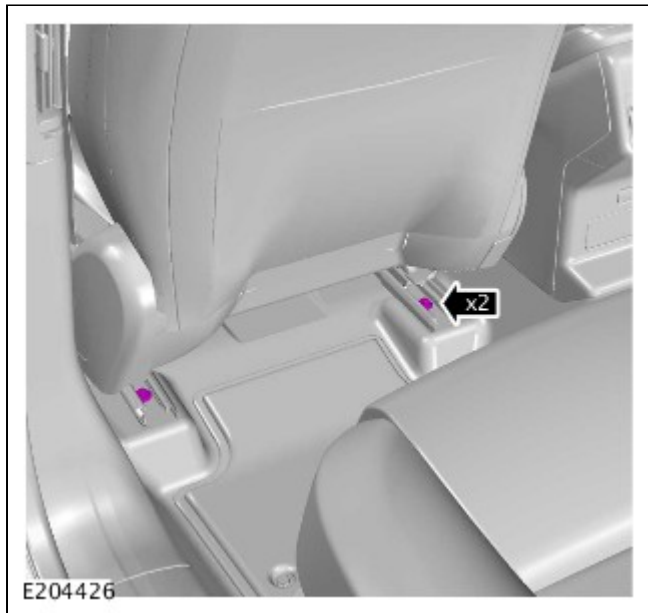
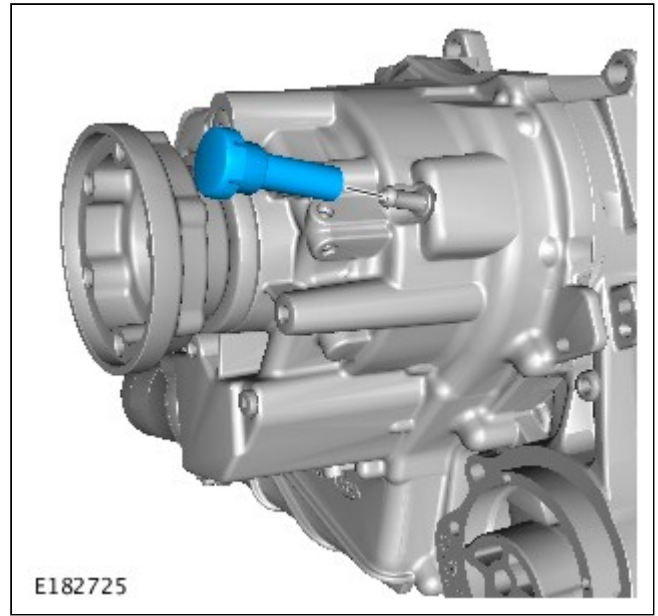
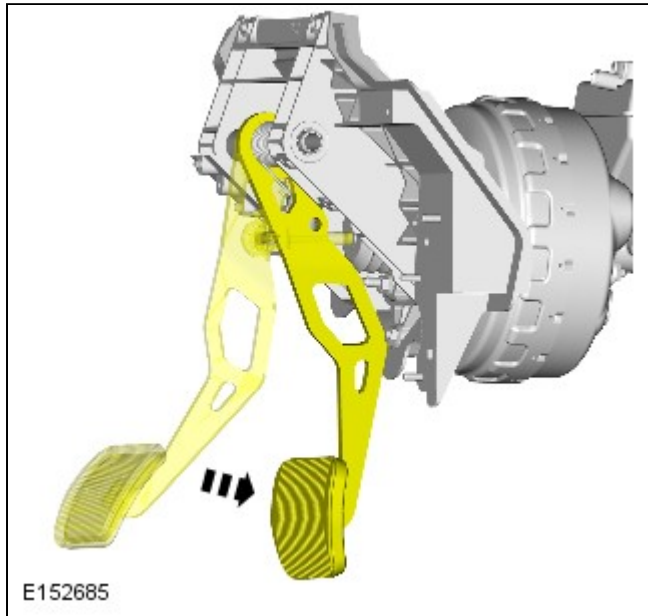
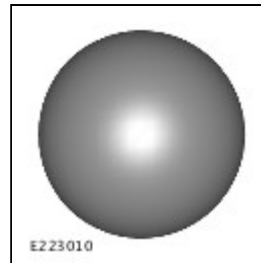
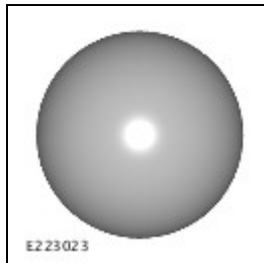
E182931

This color is used for all special tools and general equipment when already installed in previous step but they still remain part of the following steps.

2018 ONWARDS	PRE 2018
 E223031	Not Applicable
 JLR-211-343 E218896	Not Applicable
 JLR-204-726 JLR-204-836 E220379	Not Applicable

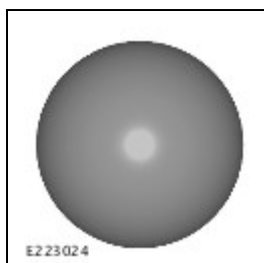
Surrounding Components

This color is used for all non-focus surrounding components.

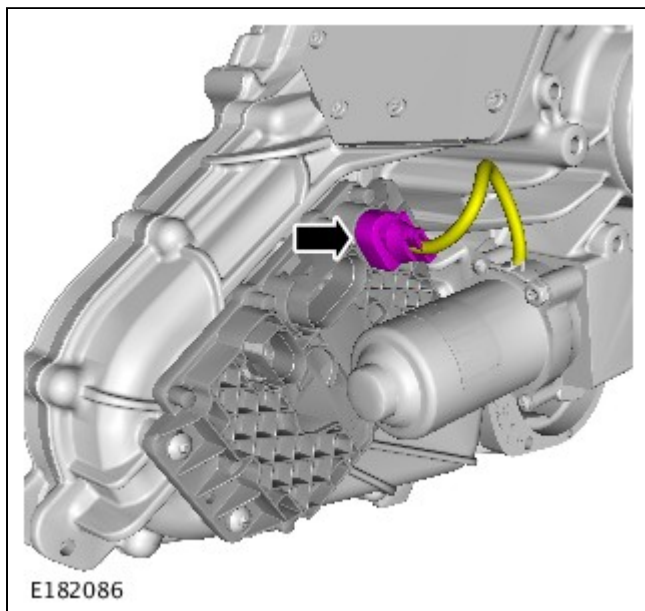


Surrounding Components - Alternative

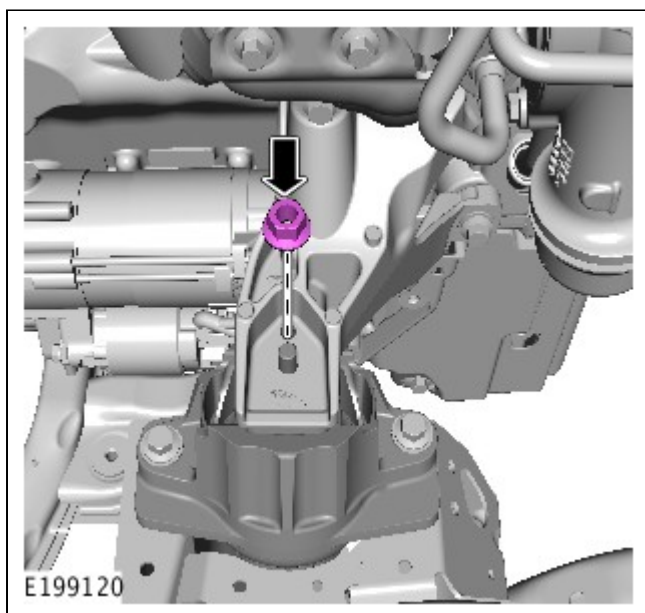
This color is can be used for non-focus components to add further clarity and depth of field.



Not Applicable



Not Applicable



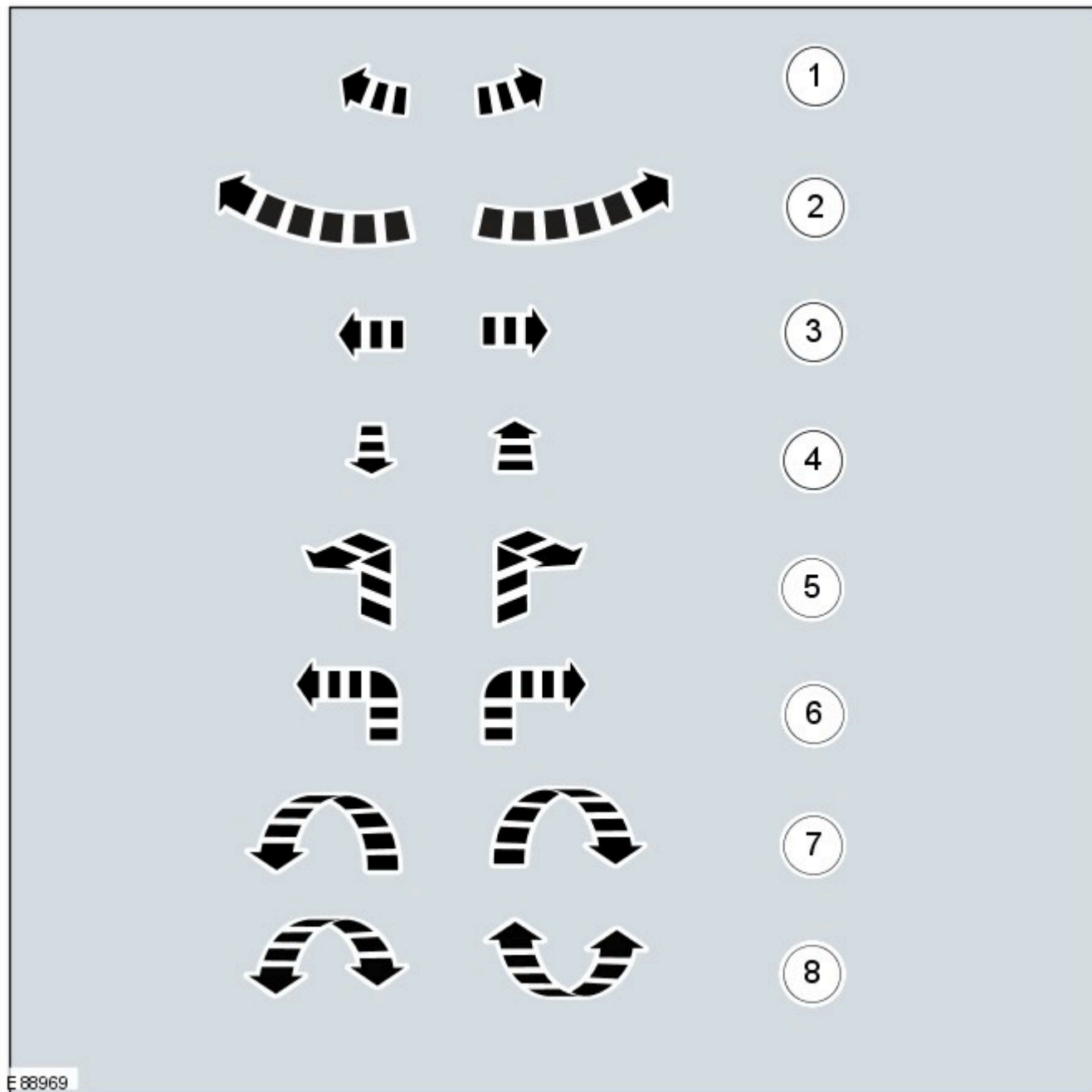
Not Applicable

ILLUSTRATION SYMBOLS

Symbols are used inside the illustrations to enhance the information display.

Movement Symbols

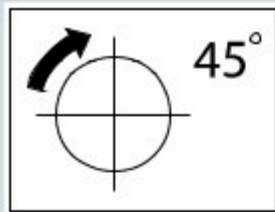
Movement symbols provide detailed information to a required component movement. These component movements can be rotational or 1-3 dimensional movements.



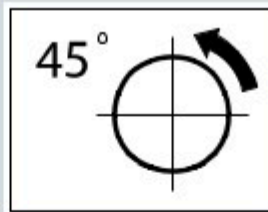
ITEM	DESCRIPTION
1	Minor component movement clockwise/counterclockwise
2	Major component movement clockwise/counterclockwise
3	Component movement to the left/right/up/down
4	Component movement towards/away
5	3 dimensional component movement
6	2 dimensional component movement
7	3 dimensional component rotation
8	3 dimensional component cycling

Turn Symbols

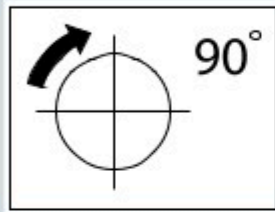
Turn symbols are used to provide further information on the direction or angle of component turns.



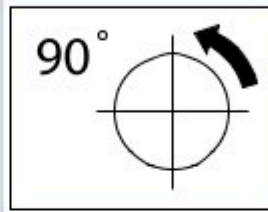
1



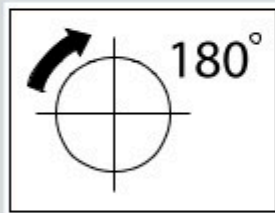
2



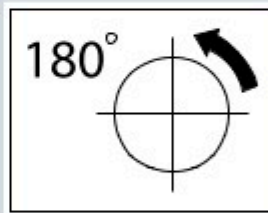
3



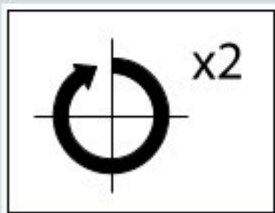
4



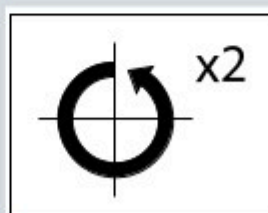
5



6



7



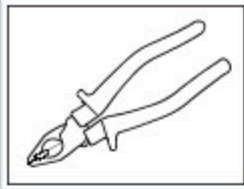
8

E 88970

ITEM	DESCRIPTION
1	Turn the component clockwise through 45°
2	Turn the component counterclockwise through 45°
3	Turn the component clockwise through 90°
4	Turn the component counterclockwise through 90°
5	Turn the component clockwise through 180°
6	Turn the component counterclockwise through 180°
7	Turn the component clockwise through 2 complete turns
8	Turn the component counterclockwise through 2 complete turns

Pliers symbols

The pliers symbols are used to show which pliers is recommended to carry out a procedure step.



1



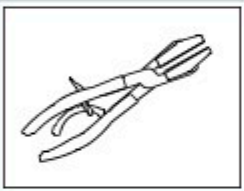
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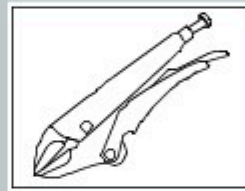
3



4



5



6



7

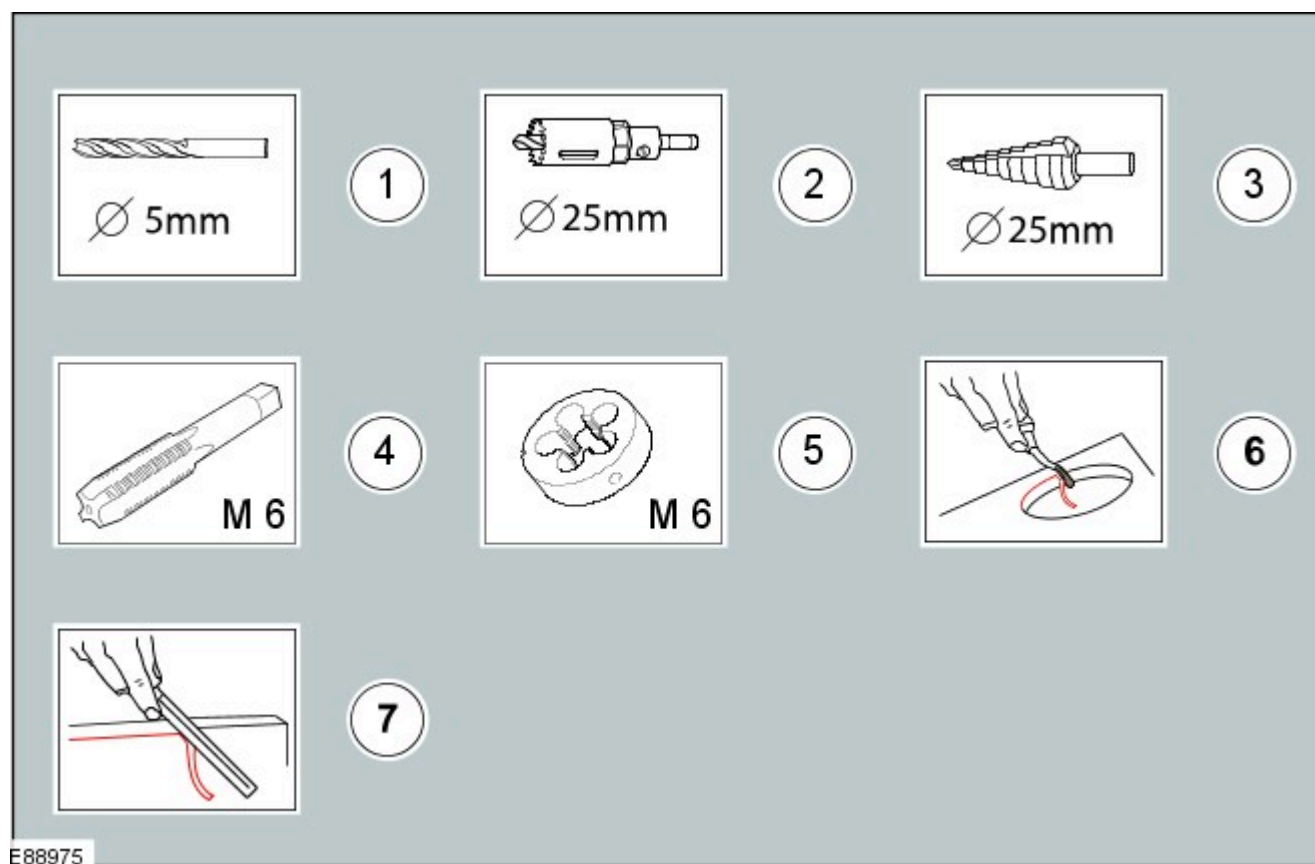
E88974

ITEM	DESCRIPTION
1	Combination pliers
2	Side cutter pliers
3	Securing ring pliers - inner
4	Securing ring pliers - outer
5	Hose clamp pliers
6	Locking pliers

7	Long nose pliers
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Drill symbols

The drill symbols are used to show which type and size of drill bit is recommended to carry out a procedure step.



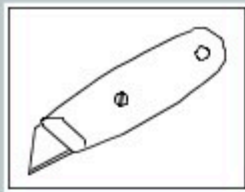
ITEM

DESCRIPTION

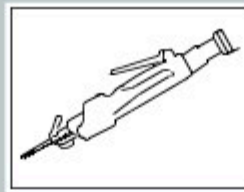
1	Drill bit with a specified diameter
2	Hole saw with a specified diameter
3	Stepped drill bit with a specified diameter
4	Tap with a specified diameter
5	Die with a specified diameter
6	Scraper for circular holes
7	Scraper for straight edges

Cutting tool symbols

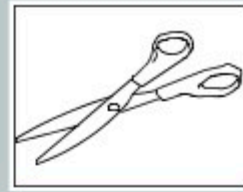
The cutting tool symbols are used to show which type of cutting tool is recommended to carry out a procedure step.



1



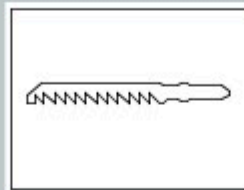
2



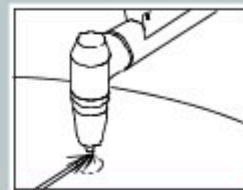
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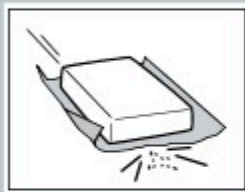
4



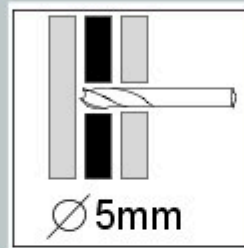
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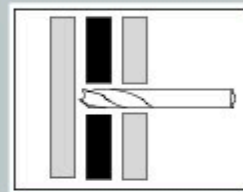
6



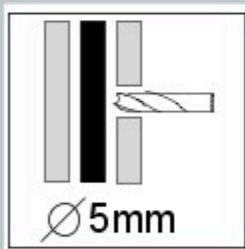
7



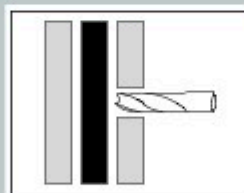
8



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12

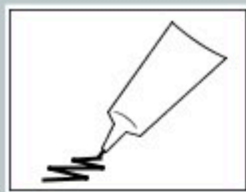
E88976

ITEM	DESCRIPTION
1	Cutting knife
2	Air body saw
3	Scissors
4	Grinder
5	Jig saw
6	Plasma cutter

ITEM	DESCRIPTION
7	Sanding Paper
8	Drill through the shown number of body panel layers with a specified diameter
9	Drill through the shown number of body panel layers with a suitable diameter
10	Drill through 1 body panel layer with a specified diameter
11	Drill through 1 body panel layer with a suitable diameter
12	Wire brush

Apply Chemical or load symbols

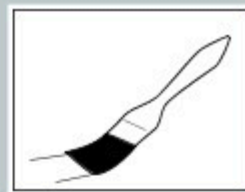
The apply chemical or load symbols are used to show where to apply which type of chemical or load to carry out a procedure step.



1



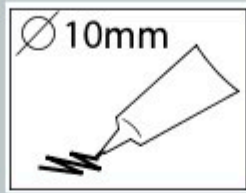
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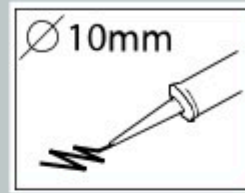
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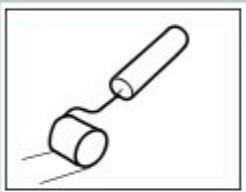
4



5



6



7



8



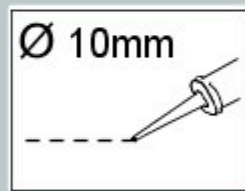
9



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11



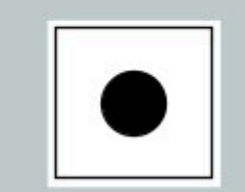
12



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18

E88977

ITEM	DESCRIPTION
1	Apply a bead from the specified tube
2	Apply a bead from the specified cartridge
3	Apply the specified chemical with a brush
4	Apply the specified load to the specified component
5	Apply a bead with a specific diameter from the specified tube
6	Apply a bead with a specific diameter from the specified cartridge

ITEM	DESCRIPTION
7	Apply the specified chemical with a roller
8	Apply hot glue to the specified component
9	Apply the specified amount of fluid from the fluid can
10	Apply fluid from the fluid can
11	Clean the specified component with the specified material
12	Apply a broken bead from the specified tube
13	Apply the specified chemical from a spray can
14	Apply the specified lubricant to the specified component
15	Apply spot welds to the specified component
16	Apply a continuous weld to the specified component
17	Handle the fluid using a syringe
18	Extract the specified amount of fluid using a syringe

Measurement symbols

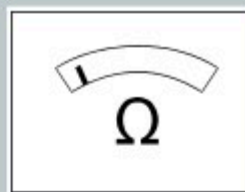
The measurement symbols are used to show where to measure which type of measurement to carry out a procedure step.



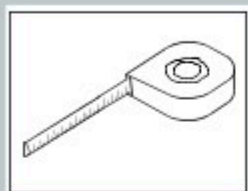
1



2



3



4



5



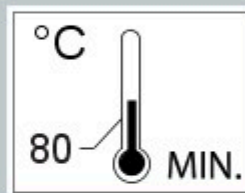
6



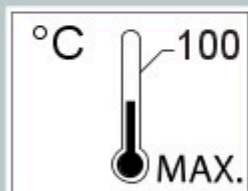
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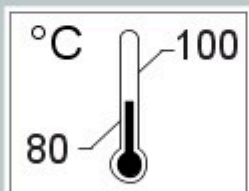
8



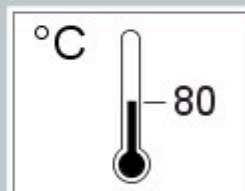
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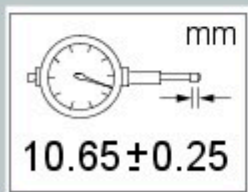
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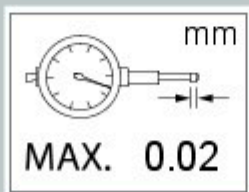
11



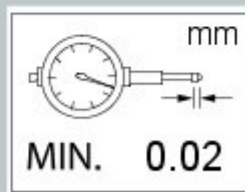
12



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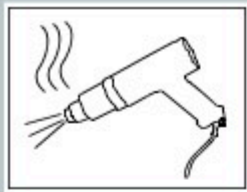
E88978

ITEM	DESCRIPTION
1	Measure the current using a digital multimeter
2	Measure the voltage using a digital multimeter
3	Measure the resistance using a digital multimeter
4	Measure the length/distance
5	Check that the specified pressure is available using a suitable pressure gauge
6	Measure the pressure at the specified port using a suitable pressure gauge

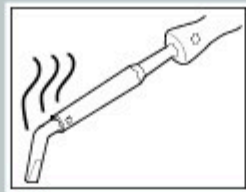
ITEM	DESCRIPTION
7	Measure the time using a suitable stopwatch
8	Wait for the specified period of time
9	The specified task requires the specified minimum temperature
10	The specified task requires the specified maximum temperature not to be exceeded
11	The specified task requires the specified temperature range
12	The specified task requires the specified temperature
13	Measure and check for the specified value using a dial indicator gauge
14	Measure and check for the specified MAX value using a dial indicator gauge
15	Measure and check for the specified MIN value using a dial indicator gauge

General equipment symbols

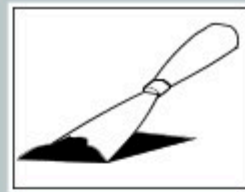
The general equipment symbols are used to show where to use which type of general equipment to carry out a procedure step.



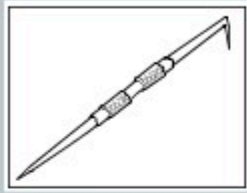
1



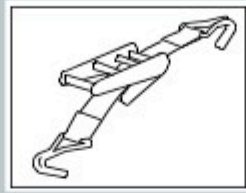
2



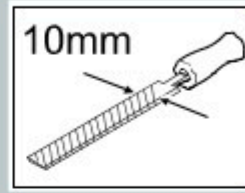
3



4



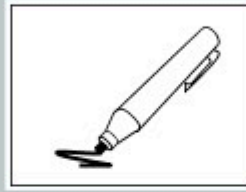
5



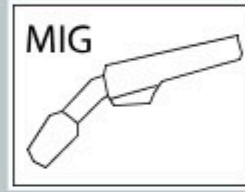
6



7



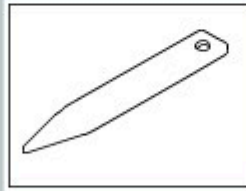
8



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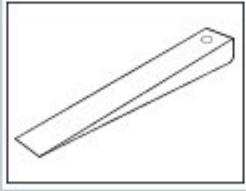
11



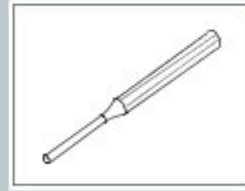
12



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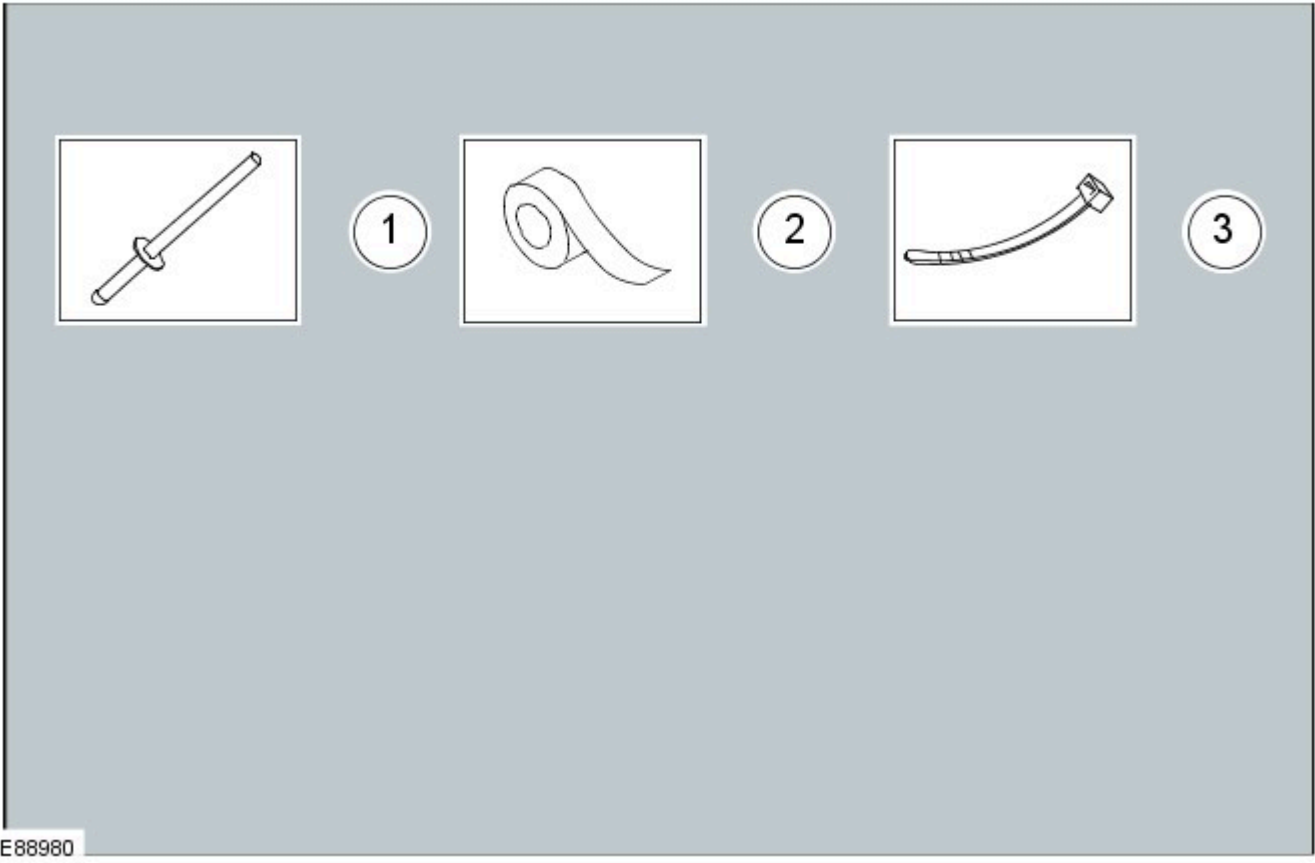
E88979

ITEM	DESCRIPTION
1	Hot air gun
2	Soldering iron
3	Scraper
4	Scriber
5	Securing strap
6	File with a specified size

ITEM	DESCRIPTION
7	Center punch
8	Marker
9	Metal inert gas (MIG) welding equipment
10	Hose clamp
11	Interior trim remover
12	Vacuum cleaner
13	Strap wrench
14	Wedge
15	Pin Punch

Material symbols

The material symbols are used to show where to use which type of material to carry out a procedure step.



ITEM	DESCRIPTION
1	Remove/Install the specified blind rivet
2	Apply tape to the specified component/area
3	Remove/Install the specified cable tie

Miscellaneous symbols

These symbols provide further information that is required to carry out a procedure step.



1



2



3



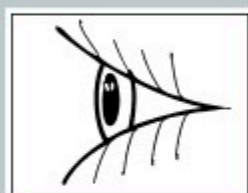
4



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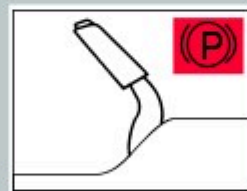
8



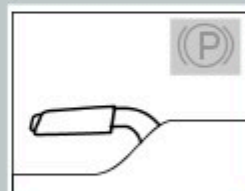
9



10



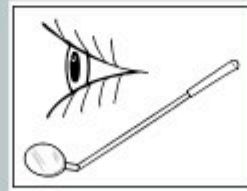
11



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16

E88981

ITEM	DESCRIPTION
1	Set the ignition switch to the 0 position
2	Set the ignition switch to the II position
3	The procedure step requires the aid of the specified number of supporting technicians
4	Self contained breathing apparatus
5	General prohibition used in combination with another symbol
6	Do not use power tools

ITEM	DESCRIPTION
7	Visual check
8	Noise check
9	Dispose the specified component
10	Replaced by item 9 (Dispose the specified component)
11	Set the engine speed to the specified value
12	Fully apply the parking brake lever
13	Fully release the parking brake lever
14	Do not dispose of batteries into the waste bin
15	Visual check using a mirror
16	Area/component must be dry

Mandatory Protective Equipment

The protective equipment symbols advise to use mandatory protective equipment to avoid, or at least, reduce possible health and safety risks.



1



2



3



4



5



6

E 89019

ITEM	DESCRIPTION
1	Wear protective gloves
2	Wear face guard
3	Wear safety goggles
4	Wear ear protectors
5	Wear safety goggles and ear protectors
6	Wear a respirator

Prohibition

The prohibition symbols are used to prohibit the specified actions to avoid or at least reduce possible component damage and health and safety risks.



1



2



3



4



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6



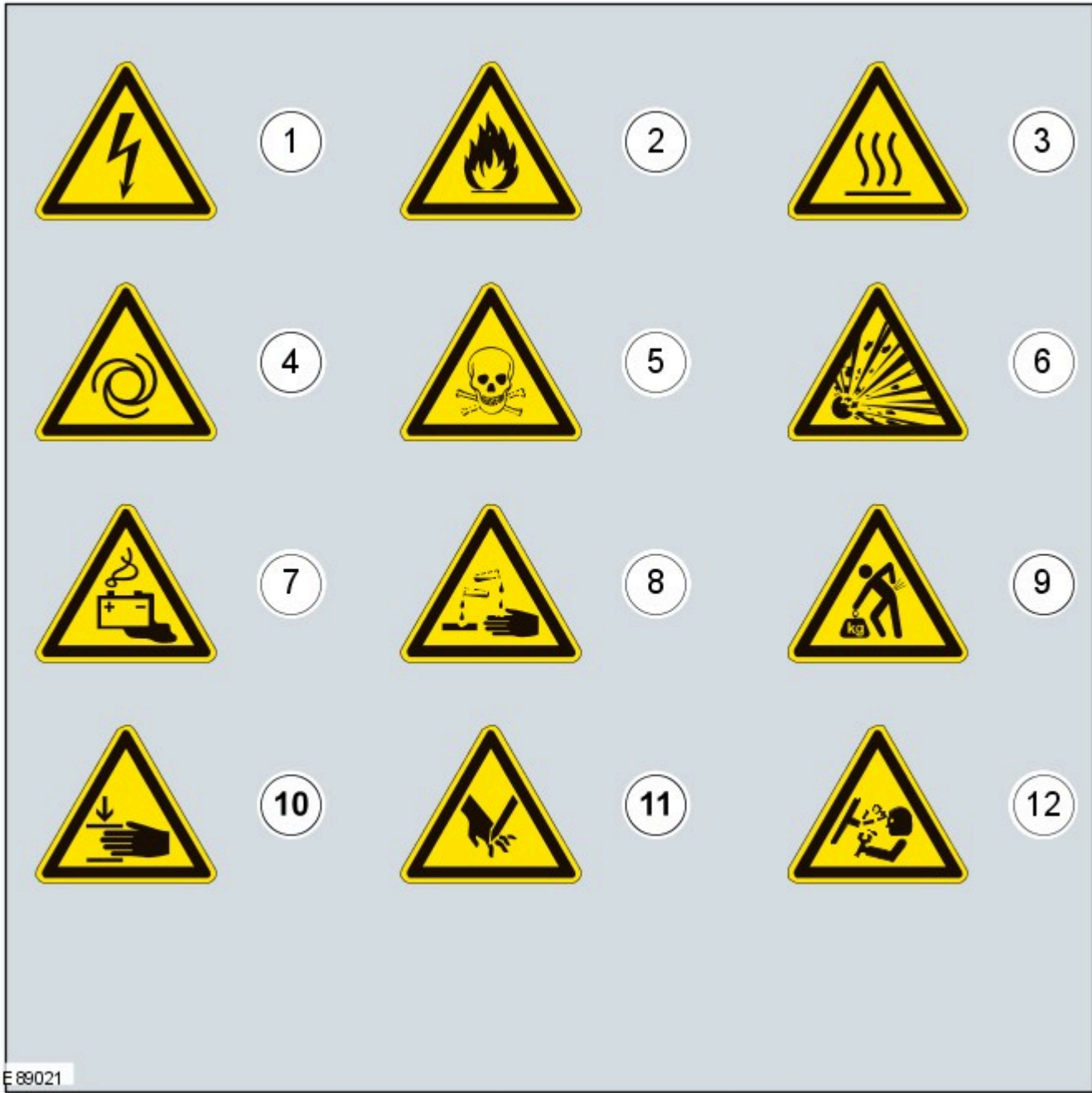
7

E89020

ITEM	DESCRIPTION
1	General prohibition symbol
2	No naked flames
3	No smoking
4	No water
5	Do not touch
6	Do not switch
7	No grinding

Warning

The warning symbols are used to advise on hazardous conditions to avoid or at least reduce possible component damage and health and safety risks.



E 89021

ITEM	DESCRIPTION
1	Hazardous voltage/Electrical shock/Electrocution
2	Fire Hazard/Highly flammable
3	Burn hazard/Hot surface
4	Automatic start-up
5	Toxic
6	Explosive material
7	Battery hazard
8	Corrosive material
9	Lifting hazard
10	Hand crush/Force from above
11	Cutting of fingers or hand

ITEM	DESCRIPTION
12	Pressure hazard

High Voltage (HV) and HV Mandatory Protective Equipment

The High Voltage (HV) symbols and HV protective equipment symbols advise to use mandatory HV protective equipment to avoid, or at least reduce, possible health and safety risks.



ITEM	DESCRIPTION
1	Power up the High Voltage (HV) system
2	Power down the High Voltage (HV) system
3	Electric Vehicle Informed Person (EVIP)
4	Electric Vehicle Competent Person (EVCP)
5	Electric Vehicle Authorised Person (EVAP)
6	Electric Vehicle Senior Authorised Person (EVSAP)

ITEM	DESCRIPTION
7	Class 0 Over Gloves
8	Class 0 Safety Glasses
9	Class 0 1000 volt, insulated rubber gloves
10	Class 1 High Voltage Safety mask
11	Class 1 CAL12.4 fire retardant lab coat
12	Safety assisting person with hook

HOW TO USE DIAGNOSIS AND TESTING PROCEDURES

Inspection and Verification

Visual Inspection Charts, Symptom Charts and other information charts (such as diagnostic routines) or supplement test procedures with technical specifications will navigate the user to a specific test procedure.

Symptom Chart

The symptom chart indicates symptoms, sources and actions to address a condition.

Pinpoint Tests

For electrical systems, pinpoint test steps are used to identify the source of a concern in a logical, step-by-step manner. Pinpoint tests have two columns: CONDITIONS and DETAILS/RESULTS/ACTIONS.

The CONDITIONS column is used exclusively for graphics and icons (with or without captions) and the DETAILS/RESULTS/ACTIONS column provides direction to another test step or specific corrective actions.

The boxed numbers indicate the order in which the described action is to be carried out.

Component Tests

A component test is used when a component is tested in multiple pinpoint tests, or if a procedure is too complicated to be formatted within a single page of the pinpoint test.

Illustrations

Test illustrations show the measurement or test to be carried out in a test step.

A representative tester graphic is used for voltmeters and ohmmeters.

If multiple measurements are made in a single graphic, the test leads are drawn with a solid line until the test lead splits to indicate the multiple measurements, at which point dashed lines are used.

Breakout box type testers are represented by a double circle test pin. Test pins are labeled with the pin number.